Sankranti Joshi

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Nationality: British | OCI Holder (India)

PROFESSIONAL SUMMARY

- Senior Machine Learning Engineer with 8+ years of experience in deep learning, computer vision, NLP, and MLOps.
- Trained vision foundation models from scratch
- Experienced in distributed training on HPC and cloud deployment
- Edinburgh MSc AI and Amazon Open Data contributor

TECHNICAL SKILLS

- Programming & Tools: Python, PyTorch, TensorFlow, Bash, Docker, PyTorch Lightning, Weights & Biases, Slurm, HPC, Poetry, scikit-learn, XGBoost, Hugging Face, DeepSpeed, Kubernetes (basic)
- Machine Learning & AI: Transformers, CNNs, RNN/LSTMs, MAE, VAE, Language Models, Contrastive Learning, Diffusion Models
- Cloud & MLOps: AWS (SageMaker, EC2, RDS), Azure, CI/CD, Model Versioning, Feature Engineering, Databricks

INDUSTRY & RESEARCH EXPERIENCE

Senior Machine Learning Engineer | PlayStation *July 2025 – Present*

ML Applied Research:

- Proposed a greenlit auto-labelling project for segmenting gameplay video, accepted into the company accelerator programme (leading 9-month project, in development) — building a shared ontology dataset to tune downstream models
- Built an inference-based object detection QA pipeline to identify characters and assets (e.g. presence/absence of specific characters, scene composition) in gameplay footage for automated quality assurance
- Developing video reasoning pipeline for gameplay videos
- Developed contextual prompts for game screen clustering using VLMs and visual prompting techniques
- Evaluated enterprise vs. self-hosted community CVAT deployment with SAM2 integration
- 3 patents submitted for filing.

Agentic AI:

- Developed a skills-based agentic workflow enabling users to set up repositories for publishing as Databricks Asset Bundles, streamlining team onboarding to Databricks
- Developed code refactor skills for Claude and Codex models to reuse code and maintain existing repository code structure as new code is added, leading to reduction in taken usage from Millions to tens of thousands.

MLOps:

- Built Databricks production pipelines for three models (menu/gameplay, HUD detection, level classification) covering training and inference at scale for computer graphics workflows
- Developed Delta Tables for streaming data collection and feature engineering
- Developed MLflow-based model serving endpoints for scalable model deployment on Databricks

Senior Machine Learning Engineer | Self-employed Contract *March 2025 – May 2025*

- Sole ML engineer for a lean health startup, building ML training and inference pipelines from scratch under limited infrastructure and noisy data conditions
- Boosted YOLO-based audio-segmentation performance by 15+ mAP points through data processing, noise cancellation and hyperparameter tuning
- Implemented reproducible training workflows on Azure ML with scalable, serverless compute
- Developed a factor analysis framework linking YOLO-based audio segmentation predictions to downstream gut health indicators, establishing an analytical approach under noisy real-world conditions
- Escalated and resolved GPU driver issues directly with Microsoft engineers, unblocking model training infrastructure

Research Student in Earth Observation Computer Vision | Sense CDT *Oct 2023 – Mar 2025*

- Developed a multimodal contextual dataset combining Landsat and Sentinel data, contributed to Amazon Open Data Sponsorship
- Designed and implemented a wavelength-conditioned input adapter for frozen Vision Transformer backbones, enabling RGB-pretrained models to be applied to multispectral Earth observation imagery
- 2.5K trainable parameters on a 112M frozen backbone; evaluated on GeoBench downstream classification
- Trained a vision foundation model from scratch using MAE-based self-supervised learning on multi-sensor satellite data, deployed on HPC and Kubernetes clusters

Applied Science Intern | Roku, Cambridge, UK *Jul 2024 – Sep 2024*

- Generated synthetic datasets using Stable Diffusion, ControlNet, and inpainting/outpainting techniques to augment YOLOX model training data
- Developed a RAG pipeline using counterfactual comparison between 3,000+ synthetically generated and real images to refine ControlNet prompts, improving synthetic data quality
- Led train/test dataset distribution strategy, improving model generalisation across real and synthetic data

Senior AIML Platform Engineer | GSK, London, UK *May 2023 – Oct 2023*

- Designed scalable ML platform pipelines on Azure HPC and cloud services, enabling researchers to deploy and run models independently across diverse use cases
- Developed an end-to-end DNA sequence classification pipeline using Hugging Face, delivering a fine-tuned model with demonstrated pruning and compression techniques — designed as both a platform demo and a reusable foundation for downstream fine-tuning

Senior Machine Learning Engineer | Informa Markets, London, UK *Sep 2022 – Mar 2023*

- Developed an LDA-guided BERT pipeline for document embeddings, improving NLP model accuracy by 25%
- Built MLOps infrastructure from scratch (data versioning, model versioning, dependency management, CI/CD), onboarding two ML scientists onto standardised workflows

Senior Associate I, Data Science | AIG, London, UK *June 2019 – Aug 2022*

- Promoted from Machine Learning Engineer to Senior Associate I, Data Science (Apr 2021)
- Built a geospatial Django platform overlaying BLS, OpenTable and multi-source datasets across US submarkets and markets (e.g. Manhattan/New York), enabling interactive visual analysis for underwriting
- Developed end-to-end feature engineering pipelines in Python with MLflow integration, enabling fully reproducible data transformations and experiment tracking
- Refactored MXNet DeepAR time-series forecasting implementation, uncovering and resolving a data leakage bug where future covariates were incorrectly used in a financial forecasting context

EDUCATION

- Master of Science in AI, The University of Edinburgh, UK (2016 – 2017)
- Master of Technology in Software Engineering, Manipal Institute of Technology, India (2012 – 2014)
- PhD Machine Learning and Geophysics, The University of Exeter (2026 – Present)